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K. J. Somaiya School of Engineering, Mumbai
Department of Artificial Intelligence & Data Science
Lab CA Miniproject Report

Historical Weather Trend & Extreme Temperature Analysis

Functional Module: Historical Meteorological Data Analysis and Extreme Event Detection

KJSCE AI Use Case: HeatWatch Climate Intelligence

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Project Links & Resources

Live Dashboard	https://oose-labca-dashboard.streamlit.app/
GitHub Repository	https://github.com/BijeetNayek0409/HeatWatch-Dashboard
Historical Dataset	India Daily Weather Dataset (Kaggle)
Live Weather API	Open-Meteo Free API

1. Aim

The aim of this project is to develop an interactive analytical dashboard that explores historical meteorological trends, identifies recurring seasonal patterns, and detects unusual temperature events using statistical anomaly detection across multiple Indian cities. It forms the historical meteorological data analysis and extreme event detection module of the HeatWatch climate intelligence use case. Specifically, the module is intended to:

- Analyse historical daily weather records (2000–2024) for ten Indian cities and present long-term yearly temperature trends and seasonal patterns through an interactive dashboard.
- Detect statistically significant temperature anomalies using a Z-score based anomaly detection algorithm computed against dynamic, city-specific and month-specific historical baselines.
- Integrate live meteorological telemetry from the Open-Meteo API and compare current conditions against the corresponding historical baseline in real time.
- Classify every observation as NORMAL, UNUSUAL or EXTREME so that heat-related risk can be identified without raising false alarms during naturally hot seasons.

2. Methodology

The module was developed as a data pipeline followed by an interactive visualisation layer. The methodology comprises the following stages.

2.1 Data Acquisition

Historical daily weather observations for the period 2000–2024 were ingested from the India Daily Weather dataset (Kaggle) for ten major Indian cities, giving a 25-year record per city. Current (live) conditions are retrieved on demand from the Open-Meteo public weather API, which provides the real-time telemetry used for comparison against historical norms.

2.2 Data Preprocessing

- Parsing of date fields and derivation of Year and Month attributes for each record.
- Cleaning of missing or invalid temperature readings so that they do not distort the baseline statistics.
- Grouping of records by city and calendar month to prepare them for baseline computation.

2.3 Trend and Seasonal Analysis

The cleaned data is aggregated to calculate yearly average maximum and minimum temperatures for each city, revealing long-term warming or cooling trends. The records are further broken down into four distinct Indian seasons – Summer, Monsoon, Post-Monsoon and Winter – to establish a localised historical baseline and make recurring seasonal patterns visible.

2.4 Baseline Computation

For every (city, month) pair, the historical mean (μ) and standard deviation (σ) of the maximum temperature are computed across all 25 years. Using a separate baseline for each month is a deliberate design decision: a temperature of 40°C is ordinary for May in many Indian cities but would be highly abnormal in January. A single annual baseline would incorrectly flag every summer day as an anomaly, whereas dynamic monthly baselines judge each reading against what is typical for that city at that time of year.

2.5 Live Telemetry Integration and Statistical Comparison

The system integrates with the Open-Meteo API to fetch real-time telemetry for the selected region. The live observation is then compared statistically against the historical baseline specifically for that city and month, which prevents seasonal bias (for example, judging a summer observation against winter averages).

2.6 Anomaly Scoring and Classification

Each observation, historical or live, is converted to a Z-score against its own city-month baseline and then classified using the thresholds in Table 1.

Absolute Z-score $ Z $	Classification	Interpretation
$ Z < 2$	NORMAL	Temperature is within the expected historical variance
$2 \leq Z < 3$	UNUSUAL	Deviates significantly from the historical average
$ Z \geq 3$	EXTREME	Mathematically extreme outlier

Table 1: Z-score classification thresholds

2.7 Dashboard Implementation

The results are presented through an interactive dashboard built with Streamlit. The dashboard is organised into three tabs:

- **Historical Analysis:** multi-year temperature trend visualisation and seasonal breakdown for the selected city over 2000–2024.
- **Live Analysis:** retrieval of current telemetry from the Open-Meteo API and comparison with the historical baseline of the selected city and month, with Z-score and a dynamic severity badge (Normal / Unusual / Extreme).
- **Extreme Events:** isolation and ranking of the top 10 most extreme historical temperature observations, each with a human-readable explanation.

2.8 Tools and Technologies

Component	Technology / Source
Programming language	Python
Data handling	Pandas, NumPy
Dashboard framework	Streamlit (deployed on Streamlit Community Cloud)
Historical dataset	India Daily Weather Dataset (Kaggle), 2000–2024
Live data source	Open-Meteo API
Version control	GitHub
Algorithm	Statistical anomaly detection (Z-score)

Table 2: Tools and technologies used

3. Algorithm

3.1 Statistical Anomaly Detection using the Z-Score Method

The Z-score expresses how many standard deviations an observation lies from the mean of its reference distribution. For a given city c and month m , the Z-score of an observed maximum temperature is defined as:

$$Z = (T_{actual} - \mu_{c,m}) / \sigma_{c,m}$$

where T_{actual} is the observed temperature, $\mu(c,m)$ is the historical mean temperature for city c in month m , and $\sigma(c,m)$ is the corresponding historical standard deviation. A positive Z indicates a temperature above the historical norm (heat anomaly) and a negative Z indicates one below it.

3.2 Procedure

1. Load the historical dataset (2000–2024) for all ten cities.
2. Extract the Year and Month from every date and remove invalid temperature records.
3. Group the data by (City, Month) and compute the mean μ and standard deviation σ of temperature for every group.
4. For each observation, look up the μ and σ of its own city and month and compute $Z = (T_{\text{actual}} - \mu) / \sigma$.
5. Classify the observation using $|Z|$: NORMAL if below 2, UNUSUAL if at least 2 but below 3, EXTREME if 3 or above.
6. For live analysis, fetch the current temperature from Open-Meteo and repeat steps 4 and 5 using the baseline of the current month.
7. Display the Z-score, classification and supporting charts on the dashboard.

3.3 Pseudocode

```
ALGORITHM HeatWatch_ZScore_Anomaly_Detection
  INPUT : D (historical records: city, date, temperature)
         T_live (current temperature), city, month
  OUTPUT: Z-score and class label

1. FOR each record in D: derive month from date
2. FOR each (city, month) group G in D:
   mu[G]      <- mean(temperature in G)
   sigma[G]   <- std_dev(temperature in G)
3. FUNCTION classify(T, city, month):
   Z <- (T - mu[city, month]) / sigma[city, month]
   IF |Z| < 2      THEN label <- "NORMAL"
   ELSE IF |Z| < 3 THEN label <- "UNUSUAL"
   ELSE           label <- "EXTREME"
   RETURN (Z, label)
4. FOR each record r in D: (Z, label) <- classify(r.temp, r.city, r.month)
5. (Z, label) <- classify(T_live, city, month)    // live comparison
```

Figure 1: Pseudocode of the Z-score anomaly detection algorithm

3.4 Justification of the Approach

- The Z-score is simple, interpretable and computationally inexpensive, which suits a dashboard that must respond in real time.
- Computing μ and σ per city and per month removes seasonality, so only departures from the local seasonal norm are flagged.
- The 2 and 3 standard-deviation thresholds follow the conventional statistical rule for identifying unusual and rare events.

4. Result

4.1 Implementation Outcome

The HeatWatch historical analysis and extreme event detection module was successfully designed and deployed as a three-tab interactive Streamlit dashboard covering ten Indian cities over the period 2000–2024. The deployed application is available at <https://oose-labca-dashboard.streamlit.app/> and its source code at <https://github.com/BijeetNayek0409/HeatWatch-Dashboard>. Table 3 summarises the delivered functionality.

Dashboard Tab	Functionality Delivered
Historical Analysis	Multi-year temperature trends and four-season breakdown per city for 2000–2024.
Live Analysis	Current Open-Meteo telemetry processed through the Z-score algorithm, with dynamic Normal / Unusual / Extreme severity badges.
Extreme Events	Top 10 most extreme historical temperature observations, ranked, with human-readable explanations.

Table 3: Summary of implemented dashboard functionality

4.2 Algorithm Performance

The Z-score algorithm successfully isolated genuine historical temperature extremes. Because each observation was compared with a dynamic baseline specific to its own city and calendar month, the algorithm did not produce false positives during naturally hot summer months. High temperatures that are routine for the season were correctly classified as NORMAL, while only readings that departed substantially from the local seasonal norm were classified as UNUSUAL or EXTREME. This proves the effectiveness of using dynamic, per-month standard deviations for climate intelligence.

4.3 Observations

- Dynamic per-month baselines proved more effective than a single annual baseline, which would have mislabelled routine summer heat as anomalous.
- The three-level classification (NORMAL, UNUSUAL, EXTREME) gives users a graded, easily interpreted indication of severity rather than a binary alert.
- The live integration with Open-Meteo allows current conditions to be assessed against 25 years of local history within the same interface.

4.4 Live Demonstration

As required by the Lab CA, a live demonstration of the implemented functional requirement was conducted. It showed the selection of a city, the yearly trend and seasonal breakdown, the live comparison with current Open-Meteo data, and the listing of detected extreme events.

4.5 Conclusion

The project demonstrates that a statistically grounded, per-city and per-month Z-score approach provides a reliable and interpretable method for detecting extreme temperature events from historical data. The module forms a dependable foundation for the wider HeatWatch climate intelligence use case, and can be extended in future with additional cities, further climate variables such as humidity and heat index, and forecasting capabilities.







